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Measurement of the Radius of
Quantum-Dot Light-Emitting Diodes
(QLEDs) by Fluorescence
Spectroscopy

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Abstract

Quantum-dot light-emitting diodes (QLEDs) exploit the size-dependent electronic structure of semiconductor nanocrystals to generate light of a well-defined colour. Because the emission wavelength of a quantum dot (QD) is governed by quantum confinement, an accurate optical characterisation of a QLED sample provides an indirect but powerful method for estimating the physical size of the emitting nanocrystals without recourse to electron microscopy. In this work, the emission spectra of five commercial QLED samples were recorded by fluorescence emission spectroscopy under a fixed excitation wavelength of 350 nm, scanning the emission range from 300 to 700 nm. The wavelength of the dominant emission peak of each sample was identified and converted into a photon energy using the relation $E [\text{eV}] = 1240/\lambda [\text{nm}]$. The particle-in-a-sphere (effective mass) approximation, originally due to Brus, was then inverted to estimate the radius of the quantum dots responsible for each emission peak. A dedicated MATLAB routine was written to automate this pipeline for an arbitrary number of samples. The five analysed QLEDs were found to emit between 544.7 and 614.2 nm, corresponding to photon energies between 2.278 and 2.020 eV and estimated radii between 0.84 and 1.16 nm. The results reproduce the expected inverse relationship between emission energy and particle radius predicted by quantum confinement theory, and they are compared with typical values reported in the literature for CdSe- and InP-based quantum dots. The limitations of the simplified model used, in particular the omission of the Coulomb correlation term, are discussed together with the associated sources of experimental uncertainty.

Keywords: quantum dots, QLED, fluorescence spectroscopy, quantum confinement, effective mass approximation, nanocrystal radius.

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Chapter 1

Introduction

1.1 Motivation

Solid-state lighting and flat-panel display technologies have undergone a profound transformation over the last two decades, driven largely by the search for emitters that combine high luminous efficiency, colour purity, and mechanical and chemical robustness. Among the various emissive technologies that have been proposed as successors or complements to organic light-emitting diodes (OLEDs), quantum-dot light-emitting diodes (QLEDs) occupy a prominent place because they combine the processability of solution-based materials with an emission colour that can be tuned continuously across the visible spectrum simply by controlling the size of the emitting nanocrystal.

The physical mechanism responsible for this tunability is quantum confinement: when the physical dimensions of a semiconductor crystal are reduced to a length scale comparable to, or smaller than, the exciton Bohr radius of the bulk material, the continuous energy bands of the bulk solid collapse into a discrete set of quantised energy levels. As a direct consequence, the effective energy gap of the nanocrystal increases as its size decreases, and the wavelength of the light that it absorbs and re-emits shifts towards the blue end of the spectrum. This effect, first rationalised theoretically by Efros and Efros and subsequently developed into a practical size-energy relation by Brus, is the physical basis of the work presented in this thesis.

From an experimental standpoint, one of the most direct and non-destructive ways of probing the size of a population of quantum dots is not by imaging them individually with an electron microscope, but by measuring the wavelength at which they emit light under optical excitation. If the emission peak of a sample can be measured with a fluorescence spectrometer, and if a reliable size-energy relation is available, the average physical radius of the quantum dots that dominate the emission of that sample can be inferred purely from an optical measurement. This indirect but experimentally convenient method is the one that has been implemented and validated in this Bachelor's

thesis.

1.2 Quantum Dots and QLED Technology

A quantum dot is a nanometre-scale crystallite of a semiconducting material, typically with a diameter between 2 and 10 nm, in which charge carriers (electrons and holes) are confined in all three spatial dimensions. Because of this three-dimensional confinement, quantum dots are often described as “artificial atoms”: like real atoms, they possess a discrete spectrum of allowed energy levels, but unlike atoms, the spacing of these levels can be engineered simply by changing the physical size of the crystallite during synthesis.

A QLED device typically consists of a thin film of quantum dots sandwiched between charge-transport layers, which are themselves placed between a cathode and an anode. Under an applied electric field, electrons and holes are injected from the opposite electrodes, transported through the charge-transport layers, and recombine radiatively inside the quantum-dot layer, producing light whose colour is set by the size of the dots. Because different-sized dots of the same material can be blended, or deposited in patterned layers, a single fabrication platform can, in principle, generate the full red-green-blue gamut required for a display, which explains the strong commercial interest in this technology for next-generation televisions and monitors.

The QLED samples studied in this thesis were not characterised electrically; instead, their optical emission was excited directly by an external ultraviolet light source in a fluorescence spectrometer, which allows the emission properties of the quantum-dot layer to be probed without needing a fully operational electroluminescent device. This is standard practice at the early stages of quantum-dot synthesis and quality control, where photoluminescence measurements are used to rapidly assess the size distribution and optical quality of a batch of nanocrystals before they are integrated into a full device stack.

1.3 Objectives of the Thesis

The specific objectives pursued in this work are the following:

- (i) To review the theoretical framework that relates the emission wavelength of a semiconductor nanocrystal to its physical radius, with particular attention to the effective mass (particle-in-a-sphere) approximation.
- (ii) To record, by fluorescence emission spectroscopy, the emission spectra of five QLED samples under a common set of instrumental conditions.

- (iii) To identify, for each sample, the wavelength and intensity of the dominant emission peak.
- (iv) To convert each measured emission wavelength into a photon energy and, subsequently, into an estimated quantum-dot radius using a simplified form of the effective mass approximation.
- (v) To implement this data-processing pipeline in a reusable MATLAB script capable of handling an arbitrary set of QLED emission peaks.
- (vi) To discuss the physical consistency of the obtained radii with the expected behaviour of quantum-confined nanocrystals, and to critically assess the limitations of the simplified model employed and the main sources of experimental uncertainty.

1.4 Structure of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 develops the theoretical background required to interpret the experimental data, starting from the electronic structure of bulk semiconductors and progressing to the particle-in-a-sphere model of a quantum-confined nanocrystal. Chapter 3 describes the experimental methodology, including the instrumentation used and the measurement protocol followed for each sample. Chapter 4 presents the experimental results: the recorded emission spectra, the identified peaks, and the radii calculated from them, together with the MATLAB implementation used to automate the calculation. Chapter 5 discusses these results in the light of quantum confinement theory, compares them with typical literature values, and analyses the sources of uncertainty and the limitations of the simplified model used. Finally, Chapter 6 summarises the main conclusions of the work and suggests possible directions for future research. A complete listing of the MATLAB code used in the analysis is provided in Appendix A.

Chapter 2

Theoretical Background

2.1 Electronic Structure of Bulk Semiconductors

In a bulk crystalline semiconductor, the periodic potential created by the regular arrangement of atoms leads, through Bloch's theorem, to the formation of continuous energy bands separated by a forbidden energy gap, or bandgap, E_g^{bulk} . Electrons occupy the lower-energy valence band, while the upper conduction band is, in an intrinsic semiconductor at zero temperature, essentially empty. When an electron is optically or electrically excited from the valence band to the conduction band, it leaves behind a positively charged vacancy, or hole, in the valence band; the bound electron-hole pair is referred to as an exciton.

Because the bands in a bulk crystal are continuous, the energy of the photon that a bulk semiconductor absorbs or emits can, to a first approximation, take any value greater than or equal to E_g^{bulk} , and the emission wavelength of a bulk sample is therefore essentially fixed by the chemical composition of the material and does not depend on the physical size of the sample (as long as the sample is macroscopic).

2.2 Quantum Confinement in Nanocrystals

The situation changes qualitatively when the physical size of the semiconductor crystallite becomes comparable to, or smaller than, the natural length scale over which the electron and hole wavefunctions extend in the bulk material – the exciton Bohr radius, a_B . For typical II-VI and III-V semiconductors used in quantum-dot synthesis, a_B lies in the range of a few nanometres, which is precisely the size regime accessible to modern colloidal synthesis techniques.

When a semiconductor crystallite is reduced to a radius $r \lesssim a_B$, the electron and hole wavefunctions become spatially confined within the volume of the nanocrystal. This spatial confinement has a direct quantum-mechanical consequence, analogous to

the well-known “particle in a box” problem: confining a particle to a finite region of space forces its momentum, and hence its kinetic energy, to increase. In the case of a quantum dot, the relevant boundary-value problem is that of a particle confined within a finite sphere of radius r , subject to (approximately) infinite potential walls at the nanocrystal surface.

The immediate physical consequence of this confinement is that the effective energy gap of the nanocrystal, $E_g(r)$, is larger than the bulk gap E_g^{bulk} , and increases monotonically as the radius r decreases. Since the emission photon energy of the nanocrystal is set by this effective gap, smaller dots emit higher-energy (shorter-wavelength, “blue-shifted”) photons, while larger dots emit lower-energy (longer-wavelength, “red-shifted”) photons. This single physical fact is what allows the emission colour of a QLED device to be tuned continuously simply by controlling the size of the constituent quantum dots during chemical synthesis.

2.3 The Effective Mass (Brus) Approximation

A widely used, and reasonably accurate, quantitative model for the size-dependence of the quantum-dot bandgap is the effective mass approximation developed by Brus, in which the electron and hole are each treated as independent particles of effective mass m_e^* and m_h^* respectively, confined within an infinite spherical potential well of radius r , and interacting through their mutual Coulomb attraction. The resulting expression for the size-dependent bandgap is

$$E_g(r) = E_g^{\text{bulk}} + \frac{\hbar^2 \pi^2}{2r^2} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right) - \frac{1.8 e^2}{4\pi \varepsilon_0 \varepsilon_r r}, \quad (2.1)$$

where $\hbar$ is the reduced Planck constant, e is the elementary charge, ε_0 is the permittivity of free space, and ε_r is the (frequency-dependent) relative dielectric constant of the semiconductor. The second term on the right-hand side of Equation (2.1) is the quantum localisation term, arising directly from the confinement of the electron and hole wavefunctions in the lowest allowed spherical mode; it scales as $1/r^2$ and dominates at small radii. The third term is the Coulomb correlation term, arising from the electrostatic attraction between the confined electron and hole; it scales only as $1/r$ and is therefore always subdominant to the confinement term in the strong-confinement regime, but it is not negligible for the radii typically encountered in colloidal quantum dots.

2.3.1 The Simplified (Confinement-Only) Model

For the purposes of a first-order estimate of the nanocrystal radius from an experimentally measured emission energy, it is common practice to neglect the Coulomb

correlation term in Equation (2.1), retaining only the quantum localisation term. Under this simplification, the size-dependent gap reduces to

$$E(r) = E_{\text{bulk}} + \frac{h^2\pi^2}{2r^2} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right), \quad (2.2)$$

which can be inverted analytically to express the radius directly in terms of the measured energy E :

$$R = \sqrt{\frac{h^2\pi^2}{2(E - E_{\text{bulk}})} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right)}. \quad (2.3)$$

Equation (2.3) is the working relation used throughout this thesis to convert measured emission energies into estimated quantum-dot radii. Its main advantage is analytical simplicity: given a single measured energy, the radius follows immediately without needing to solve a transcendental equation. Its main limitation, discussed further in Chapter 5, is that neglecting the Coulomb term systematically biases the estimated radius, typically causing it to be slightly underestimated relative to the value that would be obtained from the full Brus equation.

2.4 From Wavelength to Photon Energy

The link between the wavelength at which a quantum dot emits light and the corresponding photon energy is provided by the Planck-Einstein relation,

$$E = \frac{hc}{\lambda}, \quad (2.4)$$

where $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ is the Planck constant and $c = 3.0 \times 10^8 \text{ m/s}$ is the speed of light in vacuum. When the wavelength λ is expressed in nanometres, Equation (2.4) can be recast in the practically convenient form

$$E [\text{eV}] = \frac{1240}{\lambda [\text{nm}]}, \quad (2.5)$$

which is the relation used in this work to convert every measured emission peak wavelength into a photon energy expressed in electron-volts, prior to the radius calculation described above.

2.5 Principles of Fluorescence Emission Spectroscopy

Fluorescence (or photoluminescence) spectroscopy is an optical technique in which a sample is illuminated with light of a fixed excitation wavelength, chosen to lie above the absorption edge of the material under study, and the light subsequently re-emitted

by the sample is analysed as a function of wavelength. In a typical spectrofluorometer, the excitation light is generated by a lamp source and passed through a monochromator that selects the desired excitation wavelength; the light emitted by the sample is collected at a right angle to the excitation beam (in order to minimise the detection of directly scattered excitation light), passed through a second, scanning monochromator, and detected by a photomultiplier tube (PMT).

The output of the instrument is an emission spectrum: a plot of detected light intensity (in arbitrary or calibrated units, often referred to as “counts”) as a function of the emission wavelength. For a population of quantum dots, this emission spectrum typically consists of a single, approximately Gaussian peak centred at the wavelength corresponding to the dominant nanocrystal size present in the sample; the width of this peak reflects the size (and hence emission-energy) dispersion within the sample, a phenomenon known as inhomogeneous broadening.

2.6 Summary

The theoretical chain used in this thesis to move from a raw fluorescence spectrum to an estimated quantum-dot radius can therefore be summarised as follows: (i) a fluorescence emission spectrum is recorded for each QLED sample; (ii) the wavelength of the dominant emission peak, λ_{peak} , is identified; (iii) this wavelength is converted into a photon energy E using Equation (2.5); and (iv) this energy is inserted into Equation (2.3), together with the bulk bandgap E_{bulk} and the electron and hole effective masses of the material, to obtain the estimated radius R of the quantum dots responsible for that emission peak.

Chapter 3

Experimental Methodology

3.1 Samples

Five QLED samples, labelled QLED 1 through QLED 5, were provided for optical characterisation. The samples were nominally identical in their device architecture but were expected, based on the synthesis batches from which they were prepared, to contain quantum dots of different average size, and therefore to emit at different wavelengths across the visible spectrum. No structural characterisation (e.g. transmission electron microscopy) of the samples was performed as part of this work; the entire size characterisation reported here is based exclusively on the optical fluorescence measurements described below.

3.2 Instrumentation

All emission spectra were recorded using a bench-top spectrofluorometer with the following instrumental configuration:

- **Instrument serial number:** FS-1901001
- **Firmware version:** 140722
- **Accessory type:** Single-Cell holder
- **Photomultiplier tube (PMT) voltage:** 550 V
- **PMT integration time:** 20 ms
- **Scan speed:** 1410 nm min⁻¹
- **Scan mode:** Emission
- **Data mode:** Fluorescence

The PMT voltage of 550 V was selected to provide adequate sensitivity for the expected emission intensities of the samples while avoiding detector saturation, and the integration time of 20 ms per data point, combined with the chosen scan speed, determined the effective spectral resolution of the recorded spectra.

3.3 Measurement Protocol

For every sample, the excitation monochromator was fixed at 350 nm, a wavelength chosen because it lies well above the absorption edge of the quantum-dot materials under study, ensuring efficient above-gap excitation while remaining spectrally well separated from the visible emission band of interest. The emission monochromator was then scanned continuously from 300 to 700 nm, thereby covering both a residual signature of the excitation light (visible as a narrow peak close to 350 nm in the raw spectra) and the full visible emission band of the quantum dots. Throughout all measurements, both the excitation and emission optical paths used air as the filter medium, i.e. no additional long-pass or band-pass filters were inserted in either beam path.

Each of the five samples was measured under identical instrumental settings, which allows the recorded peak intensities to be compared with each other on a common relative scale, subject to the caveats about sample-to-sample variations in optical density and quantum yield discussed in Chapter 5.

3.4 Data Processing Overview

The raw output of the spectrofluorometer for each sample consisted of a list of (wavelength, intensity) pairs, which was exported and subsequently processed to (i) identify the wavelength and intensity of the dominant emission maximum for each sample, and (ii) feed this peak wavelength into the theoretical size-energy relation developed in Chapter 2, via the MATLAB script described in Section 4.3, in order to obtain the estimated nanocrystal radius. The following chapter presents the resulting spectra, the identified peaks, and the calculated radii.

Chapter 4

Experimental Results and Analysis

4.1 Spectral Data

The five emission spectra recorded for the QLED samples are shown together in Figure 4.1. Each curve exhibits a narrow feature near 350 nm, corresponding to residual excitation light detected by the emission monochromator, followed by a broader, approximately Gaussian emission band located between roughly 500 and 650 nm, which is the photoluminescence signal of interest arising from the quantum-dot layer of each sample.

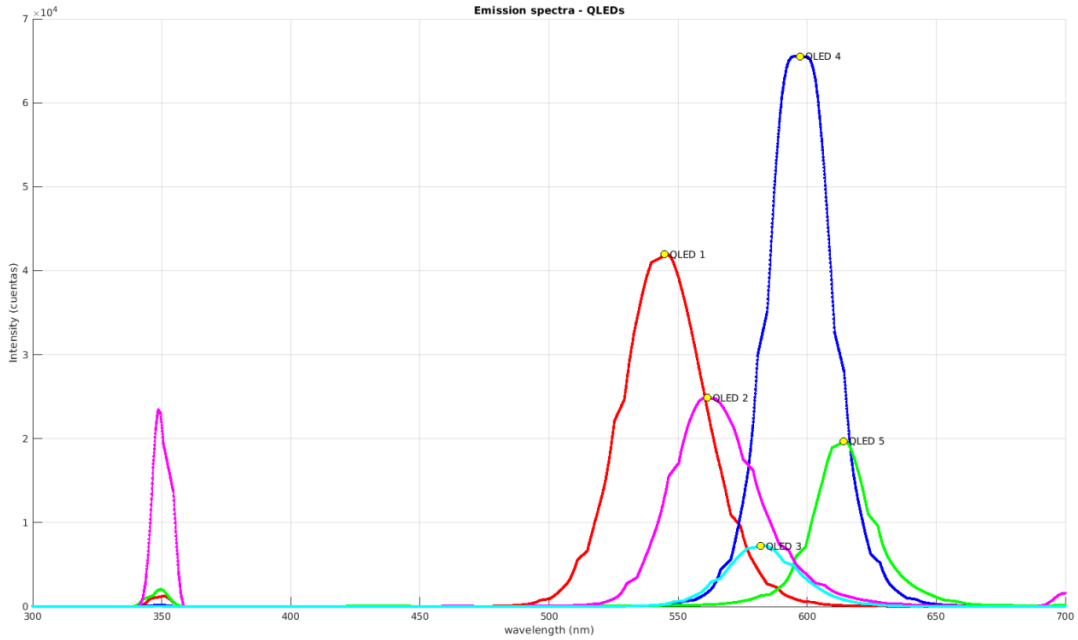


Figure 4.1: Emission spectra of the five analysed QLED samples, recorded under a fixed excitation wavelength of 350 nm. The dominant emission maximum of each sample is marked with a yellow circle.

The wavelength and intensity of the dominant emission maximum identified for each sample are summarised in Table 4.1.

Sample	λ_{peak} (nm)	Peak intensity (counts)
QLED 1	544.7	41968.0
QLED 2	561.3	24860.0
QLED 3	582.0	7209.7
QLED 4	597.2	65519.2
QLED 5	614.2	19708.9

Table 4.1: Recorded QLED emission peak wavelengths and intensities.

The five samples span a range of emission wavelengths from 544.7 nm (QLED 1, appearing green-yellow) to 614.2 nm (QLED 5, appearing orange), covering a substantial fraction of the visible spectrum accessible to quantum-dot emitters of a given base material. The peak intensities vary considerably from sample to sample, with QLED 4 showing the strongest signal (65 519.2 counts) and QLED 3 the weakest (7209.7 counts); as discussed in Section 5.3, this intensity variation most likely reflects differences in sample optical density and quantum yield rather than differences in nanocrystal size, and it is therefore not used quantitatively in the radius estimation that follows.

4.2 Calculation of the Quantum Dot Radius

As introduced in Chapter 2, the radius R of the quantum dot responsible for a given emission peak was estimated using the confinement-only (simplified) form of the effective mass approximation,

$$E(R) = E_{\text{bulk}} + \frac{h^2\pi^2}{2R^2} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right), \quad (4.1)$$

which rearranges to give the radius directly in terms of the measured photon energy:

$$R = \sqrt{\frac{h^2\pi^2}{2(E - E_{\text{bulk}})} \left(\frac{1}{m_e^*} + \frac{1}{m_h^*} \right)}. \quad (4.2)$$

For the calculations reported here, the electron and hole effective masses were taken as $m_e^* = 0.13 m_0$ and $m_h^* = 0.45 m_0$ (where m_0 is the free-electron mass), and the bulk bandgap was taken as $E_{\text{bulk}} = 1.74 \text{ eV}$, values that are representative of the semiconductor material used in the quantum-dot samples under study.

4.2.1 Computational Implementation

The energy and radius calculations described above were automated for all five samples using the MATLAB script listed in Appendix A and reproduced in abbreviated form in Listing 4.1. The script takes as input the list of measured peak wavelengths, computes

the corresponding photon energies via Equation (2.5), and inverts Equation (2.3) to obtain the estimated radius, expressed in nanometres, for every sample.

Listing 4.1: Core of the MATLAB script used for the energy and radius calculations. The full script is reproduced in Appendix A.

```

1 % Physical constants
2 h = 6.626e-34; c = 3e8; hbar = h/(2*pi);
3 q = 1.602e-19; m0 = 9.109e-31; me = 0.13*m0; mh = 0.45*m0;
4 Eg_bulk = 1.74*q;
5
6 % Wavelengths (nm)
7 lambda_nm = [544.7, 561.3, 582.0, 597.2, 614.2];
8 lambda_m = lambda_nm * 1e-9;
9
10 % Energy and radius calculation
11 E_J = (h*c) ./ lambda_m;
12 E_eV = E_J / q;
13 deltaE = E_J - Eg_bulk;
14
15 r_m = sqrt( hbar^2 * pi^2 * (1/me + 1/mh) ./ (2 * deltaE) );
16 r_nm = r_m * 1e9;

```

4.3 Final Results

Table 4.2 lists, for each of the five QLED samples, the measured emission wavelength, the corresponding photon energy calculated from Equation (2.5), and the estimated quantum-dot radius calculated from Equation (2.3).

Sample	λ (nm)	E (eV)	R (nm)
QLED 1	544.7	2.2780	0.84
QLED 2	561.3	2.2106	0.90
QLED 3	582.0	2.1320	0.98
QLED 4	597.2	2.0777	1.06
QLED 5	614.2	2.0202	1.16

Table 4.2: Calculated photon energy and estimated quantum-dot radius for each sample.

Figure 4.3 shows, in schematic form, the trend of the estimated radius as a function of the emission wavelength for the five samples; the monotonic increase of R with λ is precisely the qualitative behaviour predicted by quantum confinement theory in Chapter 2.

λ (nm)	R (nm)
544.7	0.84
561.3	0.90
582.0	0.98
597.2	1.06
614.2	1.16

Table 4.3: Estimated radius as a function of emission wavelength (data of Table 4.2, reordered for clarity).

Chapter 5

Discussion

5.1 Consistency with Quantum Confinement Theory

The central qualitative prediction of quantum confinement theory, developed in Chapter 2, is that the emission energy of a semiconductor nanocrystal should increase monotonically as its physical radius decreases. The results summarised in Table 4.2 reproduce this trend exactly: as the emission wavelength increases from 544.7 to 614.2 nm, the corresponding photon energy decreases monotonically from 2.278 to 2.020 eV, while the estimated radius increases monotonically from 0.84 to 1.16 nm. This internal consistency between the five independent measurements provides good evidence that the simplified effective-mass model of Equation (2.3), despite its approximations, captures the essential physics governing the size-dependent emission of the quantum dots under study.

Physically, this trend can be understood directly from the particle-in-a-sphere picture developed in Section 2.3: a smaller quantum dot confines the electron and hole wavefunctions more tightly, which raises the zero-point kinetic energy of both carriers, widens the effective energy gap, and therefore shifts the emission towards higher photon energies (shorter wavelengths). Conversely, larger dots impose a weaker confinement, allow the carrier wavefunctions to spread over a larger volume, and thus emit at lower energies (longer wavelengths), consistent with the calculated radii reported in Table 4.2.

5.2 Comparison with Typical Literature Values

The estimated radii obtained in this work, spanning approximately 0.84 to 1.16 nm, are somewhat smaller than the radii typically reported in the literature for CdSe-based colloidal quantum dots emitting in the same spectral window (roughly 540 to 615 nm), which are generally found to lie in the range of 1.5 to 3 nm. Several factors can

plausibly account for this discrepancy. First, the simplified model used here neglects the Coulomb correlation term of the full Brus equation (Equation (2.1)); since this term lowers the predicted energy for a given radius (or, equivalently, requires a larger radius to reproduce a given measured energy once it is included), omitting it necessarily leads to an underestimation of the true nanocrystal radius, as discussed further in Section 5.4. Second, the specific choice of effective masses and bulk bandgap used in the calculation strongly affects the absolute scale of the inferred radii, and the values adopted here, while representative, were not independently verified against the exact material composition of the samples under study. The relative trend among the five samples – i.e. the fact that radius increases monotonically with wavelength – is, however, expected to be robust to these systematic effects, since they affect all five samples in essentially the same way.

5.3 Sources of Experimental Uncertainty

Several sources of uncertainty affect the results reported in this thesis, and it is useful to distinguish between those associated with the spectroscopic measurement itself and those associated with the theoretical model used to convert the measured wavelength into a radius.

On the measurement side, the finite spectral resolution of the instrument, set by the combination of scan speed (1410 nm min^{-1}) and PMT integration time (20 ms), imposes a lower bound on the precision with which the peak wavelength of each emission band can be located; in addition, the inhomogeneous broadening of the emission peaks, arising from the natural size dispersion within each sample, means that the identified peak wavelength represents an average over a distribution of nanocrystal sizes rather than the emission of a single, perfectly monodisperse population. The considerable sample-to-sample variation in peak intensity observed in Table 4.1 (ranging from 7209.7 to 65 519.2 counts) is most plausibly attributed to differences in the optical density and photoluminescence quantum yield of the different samples, rather than to differences in nanocrystal size, and for this reason peak intensity was not used as a quantitative input to the radius calculation.

On the modelling side, as already discussed in Section 5.2, neglecting the Coulomb correlation term of the full Brus equation introduces a systematic bias into the estimated radii. In addition, the effective mass approximation itself is known to become progressively less accurate as the nanocrystal radius approaches the atomic scale, since it relies on a continuum, parabolic-band description of the electronic structure that is strictly valid only for radii sufficiently larger than the lattice constant of the material.

5.4 Limitations of the Simplified Model

It is important to be explicit about the approximations underlying the size-energy relation used in this thesis. The full effective mass approximation of Brus, given in Equation (2.1), includes both a quantum confinement term, scaling as $1/r^2$, and a Coulomb correlation term, scaling as $1/r$. In this work, only the confinement term was retained (Equation (2.2)), which allows the radius to be obtained by simple algebraic inversion (Equation (2.3)) rather than by numerically solving a transcendental equation. This simplification is a reasonable first approximation, particularly in the strong-confinement regime where the $1/r^2$ term dominates, but it systematically overestimates the true confinement energy for a given radius, and therefore systematically underestimates the true radius for a given measured energy, since the (negative) Coulomb term is entirely omitted.

A more accurate treatment would retain the full Brus equation and solve for r numerically (for instance using a root-finding algorithm such as MATLAB's `fzero`), and an even more refined treatment would go beyond the simple effective mass approximation altogether, incorporating, for example, non-parabolic band corrections or a full $\mathbf{k} \cdot \mathbf{p}$ description of the electronic structure. Such refinements were considered beyond the scope of this Bachelor's thesis, whose primary purpose was to establish and validate the basic experimental and computational pipeline linking a fluorescence emission measurement to an estimated nanocrystal radius, but they are natural directions for future work, as discussed in Chapter 6.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

In this Bachelor’s thesis, the emission spectra of five QLED samples were recorded by fluorescence emission spectroscopy under a common excitation wavelength of 350 nm, and the dominant emission peak of each sample was identified in the 300–700 nm scan range. The measured peak wavelengths were converted into photon energies using the Planck-Einstein relation, and these energies were, in turn, converted into estimated quantum-dot radii using a simplified, confinement-only form of the effective mass (Brus) approximation. This complete data-processing pipeline was implemented in a dedicated MATLAB script, capable of automatically computing the photon energy and estimated radius for any list of input emission wavelengths.

The five analysed samples were found to emit between 544.7 and 614.2 nm, corresponding to photon energies between 2.278 and 2.020 eV and estimated radii between 0.84 and 1.16 nm. The clear monotonic increase of the estimated radius with increasing emission wavelength (equivalently, the monotonic decrease of radius with increasing photon energy) reproduces the qualitative behaviour predicted by quantum confinement theory, and provides a satisfactory internal validation of the simplified effective-mass model used, notwithstanding the systematic underestimation of the absolute radius scale discussed in Chapter 5, which is attributable primarily to the omission of the Coulomb correlation term in the model.

Overall, this work demonstrates that a comparatively simple fluorescence spectroscopy measurement, combined with a lightweight analytical model and a short piece of MATLAB code, is sufficient to obtain physically consistent, order-of-magnitude estimates of the size of the quantum dots present in a QLED sample, without requiring direct structural characterisation techniques such as transmission electron microscopy.

6.2 Future Work

Several extensions of the present work would be valuable and are proposed as natural continuations of this thesis:

- **Full Brus equation.** Replacing the simplified, confinement-only radius formula with a numerical solution of the full Brus equation (including the Coulomb correlation term) would allow a more accurate, less biased estimate of the absolute nanocrystal radius to be obtained.
- **Independent structural validation.** Complementing the optical measurements with transmission electron microscopy (TEM) or small-angle X-ray scattering (SAXS) measurements of the same samples would allow the radii inferred from fluorescence spectroscopy to be directly validated against an independent, structural measurement of nanocrystal size.
- **Quantum yield and intensity analysis.** A more detailed analysis of the peak intensities recorded for each sample, normalised against a reference standard of known quantum yield, could provide additional information about the optical quality and defect density of the different QLED samples, complementing the size information extracted from the peak position.
- **Temperature-dependent measurements.** Repeating the fluorescence measurements as a function of temperature would allow the thermal broadening and shift of the emission peaks to be characterised, providing further insight into the electron-phonon coupling of the confined carriers.
- **Electroluminescence characterisation.** Extending the optical characterisation performed here (photoluminescence under external excitation) to a full electroluminescence characterisation of an operating QLED device would allow the emission properties measured under photoexcitation to be compared directly with those obtained under electrical operation, which is the relevant regime for display applications.

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Appendix A

Full MATLAB Script

The complete MATLAB script used to compute the photon energy and estimated radius for every QLED sample analysed in this thesis is reproduced below.

Listing A.1: Complete MATLAB script for radius and energy calculations.

```
1 % =====
2 % QLED Radius Estimation from Fluorescence Emission Peaks
3 % =====
4
5 % Physical constants
6 h      = 6.626e-34;      % Planck constant [J s]
7 c      = 3e8;           % Speed of light [m/s]
8 hbar   = h/(2*pi);      % Reduced Planck constant [J s]
9 q      = 1.602e-19;     % Elementary charge [C]
10 m0     = 9.109e-31;     % Free electron mass [kg]
11
12 % Effective masses (fractions of free-electron mass)
13 me     = 0.13 * m0;     % Electron effective mass
14 mh     = 0.45 * m0;     % Hole effective mass
15
16 % Bulk bandgap of the semiconductor material
17 Eg_bulk = 1.74 * q;     % [J]
18
19 % Measured emission peak wavelengths (nm)
20 lambda_nm = [544.7, 561.3, 582.0, 597.2, 614.2];
21 lambda_m  = lambda_nm * 1e-9; % Convert to metres
22
23 % -----
24 % Step 1: Photon energy from wavelength (Planck-Einstein)
25 % -----
26 E_J     = (h * c) ./ lambda_m; % Energy in joules
```



```
27 E_eV = E_J / q; % Energy in electron-volts
28
29 % -----
30 % Step 2: Radius from energy (simplified Brus equation)
31 % -----
32 deltaE = E_J - Eg_bulk;
33 r_m = sqrt( hbar^2 * pi^2 * (1/me + 1/mh) ./ (2 * deltaE) );
34 r_nm = r_m * 1e9; % Convert to nanometres
35
36 % -----
37 % Display results
38 % -----
39 fprintf('Wavelength (nm)\tEnergy (eV)\tRadius (nm)\n');
40 for i = 1:length(lambda_nm)
41     fprintf('%0.1f\t\t%0.4f\t\t%0.2f\n', lambda_nm(i), E_eV(i),
42         r_nm(i));
43 end
```